121. Best Time to Buy and Sell Stock

You are given an array prices where prices[i] is the price of a given stock on the i^{th} day.
You want to maximize your profit by choosing a **single day** to buy one stock and choosing a **different day in the future** to sell that stock.
Return *the maximum profit you can achieve from this transaction*. If you cannot achieve any profit, return 0.

Example 1:
Input: prices = [7,1,5,3,6,4]
Output: 5
Explanation: Buy on day 2 (price = 1) and sell on day 5 (price = 6), profit = 6-1 = 5.
Note that buying on day 2 and selling on day 1 is not allowed because you must buy before you sell.
Example 2:
Input: prices = [7,6,4,3,1]
Output: 0
Explanation: In this case, no transactions are done and the max profit = 0.

So in this questions lets understand problem statement

You are given an array prices where:

- prices[i] = stock price on day i

You must choose:

- **one day to buy**
- **one later day to sell**

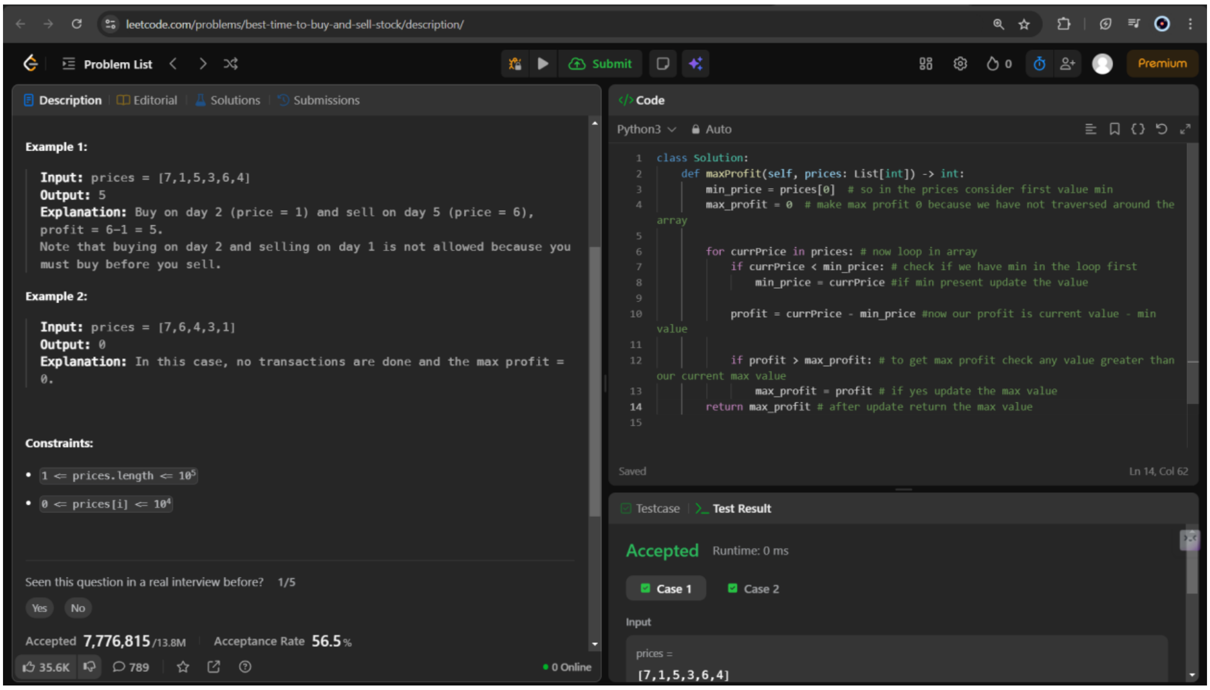
Return the **maximum profit** you can make.
If no profit is possible, return 0.

Main idea
You do **not** need to check every pair.
Instead, while moving from left to right:

- keep track of the **lowest price seen so far**
- at each day, calculate:

profit = current_price - minimum_price_so_far

- update the maximum profit


leetcode.com/problems/best-time-to-buy-and-sell-stock/description/
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Example 2:
Input: prices = [7,6,4,3,1]
Output: 0
Explanation: In this case, no transactions are done and the max profit = 0.
Constraints:
1 <= prices.length <= 10^5
0 <= prices[i] <= 10^4
Seen this question in a real interview before? 1/5
Yes
No
Accepted 7,776,815 /13.8M
Acceptance Rate 56.5 %
35.6K
789
0 Online
Code
Python3
Auto
class Solution:
def maxProfit(self, prices: List[int]) -> int:
min_price = prices[0] # so in the prices consider first value min
max_profit = 0 # make max profit 0 because we have not traversed around the array
for currPrice in prices: # now loop in array
if currPrice < min_price: # check if we have min in the loop first
min_price = currPrice #if min present update the value
profit = currPrice - min_price #now our profit is current value - min value
if profit > max_profit: # to get max profit check any value greater than our current max value
max_profit = profit # if yes update the max value
return max_profit # after update return the max value
Saved
Ln 14, Col 62
Testcase
Test Result
Accepted
Runtime: 0 ms
Case 1
Case 2
Input
prices =
[7,1,5,3,6,4]